

Regularisation ambiguities in first-order gauged matrix models: level shifts and the Procesi bundle

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Abstract

First-order gauged matrix quantum mechanics on S^1 computes equivariant indices of line bundles on the Higgs branch, but the identification of the Chern–Simons level depends on how the theory is regularised. Different papers report different “quantum shifts”: $k' \rightarrow k' - N$ for the ADHM model of the 4d quantum Hall effect [1], $k + n \mapsto k$ for the Chern–Simons matrix model on the framed Jordan quiver [2], and $k \rightarrow k + 1$ for the Polychronakos matrix model [3]. We give a single rule that accounts for all of them. Write the one-loop factor of every field mode of weight x and gauge fugacity w as $x^{s_\varepsilon} w^{s_w} / (1 - xw)$. Then the partition function is $Z = (\prod x^{s_\varepsilon}) \chi(M, \mathcal{O}(k' - \sum \text{charge} \cdot s_w))$, where M is the Higgs branch. It equals the twisted index $\chi(M, \mathcal{O}(k') \otimes K^s)$ if and only if $s_\varepsilon = s_w = s$ for all fields. Zeta-function regularisation ($s = \frac{1}{2}$) gives the Dirac (spin^c) index; gauge-covariant time lattices have $s_w = 0$ and never shift the level. The first two literature shifts are, as their authors already noted, the same normal-ordering mechanism ($s_w = 1$ on fundamental fields). The third has been explained in the literature in three different ways: as normal ordering of the constraint, as the adjoint zero-point energy, and as a Jacobian. In the rule it is unambiguously an s_ε effect, namely the adjoint zero-point energy absorbed into the Calogero coupling, and not a Gauss-law effect.

We test the rule on three phase spaces: $\text{Hilb}^n(\mathbb{C}^2)$ (where K is trivial), the Grassmannian (where K is non-trivial and the space is compact) and the Jordan quiver variety (where K is non-trivial and the space is non-compact). The tests use Molien integrals, Jeffrey–Kirwan residues, Atiyah–Bott sums and, for gauge group $U(n)$ with $n \leq 3$, explicit time lattices. We also show where the rule stops applying: once the Hamiltonian is quadratic in a generator, the continuum coherent-state path integral distorts the spectrum, and the time lattice converges only if the Fock space is truncated first.

Finally, we lift the ADHM model to the S_n -equivariant character of the Procesi bundle. We find a further regularisation ambiguity: imposing the isospectral constraint scheme-theoretically, on the reduced scheme, or derivedly gives characters that agree with $\chi(\mathcal{O}(k))$ on the S_n -invariant part but differ elsewhere, and only the reduced prescription gives the Procesi bundle. For flag (parabolic) versions of the model we conjecture a closed formula for the virtual count at every level k and check it for $n \leq 5$.

1 Introduction

A first-order action $S = \oint \theta - \oint H dt$ with $d\theta = \omega$ defines a path-integral version of geometric quantisation. When H is the moment map of a torus action, the partition function $Z = \text{Tr } e^{-\beta H}$ on a circle is an equivariant index and can be written as a fixed-point sum. Gauged versions of this set-up appear in several places:

- the ADHM matrix model of the 4d quantum Hall effect [1], whose Higgs branch is the instanton moduli space ($\text{Hilb}^n(\mathbb{C}^2)$ in rank one);

*Claude Opus 5.5, an AI system developed by Anthropic. Almost all of this work, from the computations to the text, was generated by the AI; hence the first authorship (which alphabetical order happens to confirm).

- the Chern–Simons matrix model on the framed Jordan quiver [2, 4];
- the Polychronakos matrix model of the quantum Hall effect [3].

In each of these papers the relation between the bare Chern–Simons level and the line bundle that is actually quantised involves a shift. A refined version of the same question arises when the ADHM model is lifted to the S_n -equivariant character of the Procesi bundle, whose components are the modified Macdonald polynomials; there the ambiguity is not a level shift but the way a constraint is imposed.

What is already known. The shifts $k' \rightarrow k' - N$ [1] and $k + n \mapsto k$ [2] are both attributed by their authors to normal ordering of the fundamental-field bilinear in the trace of the Gauss law. The same “ordering issue” already appears in [4, 5], so the identification of these two shifts is not new. The shift $k \rightarrow k + 1$ of [3], which Polychronakos calls “a quantum reordering effect”, has been explained in three different ways:

- as the zero-point energy $N^2/2$ of the adjoint oscillators [6];
- as the comparison of ground-state energies, or equivalently a Vandermonde factor, between the matrix model and the Calogero model [7], and as a Jacobian (footnote in [1]);
- as “normal ordering a constraint” [8].

In addition, [1] lists it together with the Gauss-law shift. Two related questions are whether the expected half-integer ($K^{1/2}$ or spin^c) shifts appear [9–13], and whether the continuum coherent-state path integral can be trusted [14–18].

This paper organises these effects into one dictionary. Our main statements are:

1. *A rule* (Proposition 1). Each field carries two regularisation exponents, s_ε for its zero-point energy and s_w for its contribution to the Gauss law. The partition function is an overall factor times the normal-ordered index at a shifted level.
2. *A lemma* (Lemma 3). At each Jeffrey–Kirwan pole, the product of the weights of all matter modes equals the product of the tangent weights. Hence uniform symmetric regularisation computes $\chi(M, \mathcal{O}(k') \otimes K^{1/2})$.
3. *A no-go statement* (Proposition 2). Gauge-covariant time lattices have $s_w = 0$, so they never shift the level.
4. *A dictionary of the known shifts* (Table 1). The shifts of [1] and [2] are recovered as $s_w = 1$ on fundamental fields, in agreement with their authors. The shift of [3] is $s_\varepsilon = \frac{1}{2}$ on the adjoint: its Gauss law is normal ordered ($s_w = 0$), and the shift comes from the adjoint zero-point energy, whose off-diagonal part the Calogero coupling absorbs. This sides with [6, 7] and shows that attributing it to the ordering of the constraint [8], or grouping it with the Gauss-law shift [1], mixes up two different exponents.
5. *The limits of the rule* (Section 6). For Hamiltonians that are quadratic in a generator, the continuum path integral distorts the spectrum, and the time lattice has non-commuting limits.
6. *The Procesi bundle* (Section 7). Adding the joint eigenvalues as fields, the scheme-theoretic, reduced and derived impositions of the isospectral constraint give different S_n -equivariant characters, all with the same invariant part $\chi(\mathcal{O}(k))$. For flag versions of the model we conjecture a closed formula (9) for the virtual count at every level.

Most individual entries are standard, and some identifications (item 4 and part of the Calogero discussion) are already in the literature, as summarised above. What we did not find in the literature, within the scope of our search, are:

- the two-exponent rule together with the criterion $s_\varepsilon = s_w = s$ for a K^s interpretation;
- the statement that gauge-covariant time lattices never shift the level;
- the JK-pole argument for the $K^{1/2}$ twist;
- the observation that on phase spaces with non-trivial K the midpoint lattice does not reproduce the half-form shift of the level;
- the non-commuting limits of the time lattice for non-linear Hamiltonians;
- the comparison of the isospectral-constraint prescriptions, and the level- k formula (9) for virtual counts of nested Hilbert schemes (the case $k = 0$ follows from [19]).

The mathematical framework of Section 7 (Haiman’s theorems, Ginzburg’s isomorphism, the virtual-dimension criterion for flags) is known, as summarised there.

Organisation. Section 2 sets up the models and the regularisation data, and Section 3 states and proves the rule. Section 4 applies it to the three phase spaces, Section 5 collects the dictionary, and Section 6 discusses the limits of the rule. Section 7 lifts the ADHM model to the Procesi bundle. Appendix A contains remarks on JK residues and wall crossing, and Appendix B lists the ranges of the numerical checks.

The Jeffrey–Kirwan (JK) residue machinery [20–22] is used throughout, mostly as a tool. Its application to the rank-one ADHM model (Appendix A) adds only minor points to the general results of [20, 22]; see also [23].

2 Set-up

2.1 Models

Let $G = U(n)$ with gauge field A_t and bare Chern–Simons level k' , and let the matter consist of complex fields in representations of G with symplectic form $i d\bar{\phi} \wedge d\phi$. The action is

$$S = \int dt \left[\sum_{\text{fields}} i \bar{\phi} D_t \phi - k' \operatorname{tr} A_t - H \right], \quad (1)$$

where H is the moment map of a torus T (a linear combination of number operators). The Gauss law is $\mu_{\mathbb{R}} = k' \mathbf{1}$, so the FI parameter equals the bare level, $\zeta = k'$. A holomorphic constraint $\mu_{\mathbb{C}} = 0$, if present, is imposed by Koszul fermions χ with $Q = \sum (\mu_{\mathbb{C}})_{ab} \chi_{ba}$, $Q^2 = 0$. The partition function on S^1 with periodic boundary conditions reduces, after static diagonal gauge fixing, to a torus integral with Haar measure $\prod_{a \neq b} (1 - w_a/w_b)$. We consider three examples.

- **ADHM, rank one:** adjoints B_1, B_2 (weights t, q), fundamental I and antifundamental J , and $\mu_{\mathbb{C}} = [B_1, B_2] + IJ$. The Higgs branch is $M = \operatorname{Hilb}^n(\mathbb{C}^2)$. The scheme $\mu_{\mathbb{C}}^{-1}(0)$ is a reduced complete intersection [24], so imposing $\mu_{\mathbb{C}} = 0$ by Koszul fermions or by restricting to the variety gives the same character. We use the conventions of [25]: $T_\mu = t^{n(\mu)} q^{n(\mu')}$, $\mathcal{O}(1) = \wedge^n \mathcal{B}$, and the tangent weights $t^{1+l} q^{-a}$, $t^{-l} q^{1+a}$.
- **Grassmannian:** N fundamentals I_i with weights x_i and nothing else. Then $M = \operatorname{Gr}(n, N)$ and $K_M = \mathcal{O}(-N)$.
- **Framed Jordan quiver:** one adjoint Z (weight q) and f fundamentals (weights a_i). Then $M \cong \operatorname{Quot}^n(\mathcal{O}_{\mathbb{A}^1}^{\oplus f})$ [2].

2.2 Regularisation data

A single bosonic mode of weight $x = e^{-\beta\varepsilon}$ and holonomy w has the one-loop factor $1/\det(\partial_\tau + \varepsilon - ia)$. Different regularisations give $x^{s_\varepsilon} w^{s_w}/(1 - xw)$:

- normal ordering (equivalently, the forward Q-symbol time lattice) gives $s_\varepsilon = s_w = 0$;
- zeta-function regularisation with symmetric Matsubara cut-off gives $s_\varepsilon = s_w = \frac{1}{2}$;
- anti-normal ordering gives $s_\varepsilon = s_w = 1$;
- time lattices give $s_w = 0$ with $s_\varepsilon \in \{0, \frac{1}{2}, 1\}$ (Proposition 2).

Fermions contribute the inverse factor. In operator language, s_w is the ordering of the matter bilinear in the Gauss law, and s_ε is the ordering of H .

3 The rule

Proposition 1 (Rule). *Let $\delta = \sum_f c_f s_{w,f}$ be the sum, over charged fields, of the $U(1) \subset U(n)$ charge per colour c_f times s_w . Let $\mathcal{X} = \prod_{\text{modes}} x^{s_\varepsilon}$, with fermions contributing x^{-s_ε} . Then*

$$Z_{\text{reg}}(k') = \mathcal{X} \cdot Z_{\text{normal}}(k' - \delta). \quad (2)$$

Proof. For adjoint fields the factors w^{s_w} cancel between (a, b) and (b, a) . For a field of charge c_f they give $\prod_a w_a^{c_f s_w}$, which combines with the Chern–Simons factor $\prod_a w_a^{-k'}$. What remains is the x^{s_ε} factors, which are independent of w . $\square$

Proposition 2 (Gauge-covariant lattices). *Place the links U_j on the hopping terms, $\bar{z}_j(z_j - U_j z_{j-1})$, and evaluate H by any rule that is local in time and gauge covariant. Then the lattice determinant of a field in representation R is $\prod_\omega (a^M - \omega b^M)$, where ω runs over the eigenvalues of the holonomy in R . Hence $s_w = 0$ and the lattice never shifts the level. In the continuum limit, the forward Q-symbol, midpoint and same-time P-symbol prescriptions give $s_\varepsilon = 0, \frac{1}{2}, 1$ respectively.*

Proof. The kinetic matrix is block-cyclic with diagonal $a\mathbf{1}$ and sub-diagonal $-bR(U_j)$. Its determinant depends only on the conjugacy class of $R(U_M \cdots U_1)$. The values of a and b follow from expanding to first order in $\delta = \beta/M$. $\square$

Lemma 3 (Zeta function = $K^{1/2}$). *At every isolated JK pole, the product of the weights xw^ρ of all matter modes equals the product of the tangent weights of M at the corresponding fixed point. The reason is that the tangent character is the matter character minus the gauge character, and the gauge weights w_a/w_b multiply to one. Consequently, uniform zeta-function regularisation gives*

$$Z_\zeta(k') = \chi(M, \mathcal{O}(k') \otimes K_M^{1/2}). \quad (3)$$

More generally, Z_{reg} equals $\chi(M, \mathcal{O}(k') \otimes K_M^s)$ for all k' if and only if $s_\varepsilon = s_w = s$ for all fields.

Remark 4 (Global anomaly). If $\delta \notin \mathbb{Z}$, for example with an odd number of fundamentals and no antifundamentals in zeta-function regularisation, then Z_ζ is gauge invariant only for half-integer k' . This is the bosonic analogue of the fermionic global anomaly discussed in [20]. Lattices with integer level cannot reach the Dirac index in this case.

4 Three phase spaces

4.1 $\text{Hilb}^n(\mathbb{C}^2)$: trivial K

Because the field content contains both I and J , the level shift in zeta-function regularisation is $\delta = n(s_{w,I} - s_{w,J}) = 0$. Equivariantly $K \cong \mathcal{O} \otimes (qt)^n$, so the $K^{1/2}$ twist is only an overall factor $(qt)^{n/2}$, coming from the zero-point energy $E_0 = \frac{1}{2}(\sum_{\text{bos}} \varepsilon - \sum_{\text{ferm}} \varepsilon)$. The midpoint lattice $(s_w = 0, s_\varepsilon = \frac{1}{2})$ and the zeta function $(s_w = s_\varepsilon = \frac{1}{2})$ therefore agree on Hilb^n . The ordering in [1] (and the “ordering issue” of [5]) treats I anti-normally and J normally ($s_{w,I} = 1, s_{w,J} = 0$), which gives $k = k' - N$. This is not a power of K . If bosons and Koszul fermions, or B_1 and B_2 , are regularised differently, the overall factors are not powers of K either.

We checked the baseline $Z_{\text{normal}} = \chi(\text{Hilb}^n, \mathcal{O}(k))$ in four independent ways for $n \leq 4$, $0 \leq k \leq 3$:

- the Molien integral of [1];
- JK residues, where each colour-partition residue equals the corresponding Atiyah–Bott term (checked for $n \leq 5$);
- the Atiyah–Bott sum;
- the Hilbert series of products of alternants [26].

For gauge-covariant lattices with $n = 2, 3$, we computed every determinant numerically (random $U(n)$ links for the invariance check, and holonomy integrals for Z). The results converge as $O(1/M)$ to $\mathcal{X} \cdot Z_{\text{normal}}(k')$ with no level shift.

4.2 $\text{Gr}(n, N)$: non-trivial K , compact

Here $\delta = Ns_w$, and the rule gives:

- normal ordering: $s_{(k'^n)}(x) = \chi(\mathcal{O}(k'))$;
- zeta function: $(\det x)^{n/2} s_{((k' - N/2)^n)}(x) = \chi(\mathcal{O}(k') \otimes K^{1/2})$, a level shift by $N/2$;
- anti-normal ordering: $\chi(\mathcal{O}(k') \otimes K)$, a level shift by N ;
- every gauge-covariant lattice, including the midpoint rule: $(\det x)^{sn} s_{(k'^n)}$.

The naive identification “midpoint rule = Weyl ordering = metaplectic correction” is therefore correct for the zero-point part and wrong for the level part. On Hilb^n the failure is invisible because K is trivial. This does not contradict the semiclassical statements that the Weyl symbol removes the Solari–Kochetov correction [9], or that the metaplectic correction is tied to the correct continuum limit of coherent-state path integrals [10]. Those statements concern the continuum or the semiclassical propagator. Ours concerns the Chern–Simons level in a gauge-covariant time discretisation, where the half-form shift must be put in by hand through the bare level. For half-form and metaplectic-c quantisation with torus actions see [11, 12]; in the quantum-Hall context the metaplectic correction is sometimes deliberately omitted [13]. We checked this for $(n, N) \in \{(1, 2), (1, 3), (2, 3), (2, 4)\}$ and $k \leq 3$.

4.3 Jordan quiver: non-trivial K , non-compact

In normal ordering the Molien integral and the JK residues reproduce Theorem A of [2] (which notes that the formula first appeared in [4]),

$$\text{ch } \mathcal{H} = \tilde{H}_{(k^n)}(a; q) \prod_{i=1}^n \frac{1}{1 - q^i}, \quad (4)$$

where $\tilde{H}_\mu$ is built from Jing operators as in that reference. We checked this for $f \leq 2$, $n \leq 3$ and $k \leq 3$. In zeta-function regularisation the JK residues multiplied by (product of tangent weights) $^{1/2}$ reproduce Z_ζ , which confirms Lemma 3 here with $K \cong \mathcal{L}_{\det}^{-f}$ and a level shift of $f/2$. The shift $k + n \mapsto k$ of [2] is $s_w = 1$ on fundamentals. For $f = 1$, the Calogero model obtained from the Hermitian (real-polarised) version [3] with coupling $\ell(\ell - 1)$, $\ell = k + 1$, satisfies

$$Z_{\text{Cal}}(\ell = k + 1) = q^{n^2/2} a^{-kn} Z_{\text{normal}}(k), \quad (5)$$

which we checked for $n = 2$ by finite differences. The factor $q^{n^2/2}$ is the zero-point energy of the adjoint ($s_\varepsilon = \frac{1}{2}$). Its off-diagonal part $q^{n(n-1)/2}$ is the degree of the Vandermonde factor, which the Calogero coupling absorbs. Thus the shift of [3] is an s_ε effect, not an s_w effect. The relation to the zero-point energy is not new: [6] identify the $N^2/2$ zero-point term as the cause of the shift, and [7] compare the ground-state energies $N^2/2$ and $N/2$ of the matrix and Calogero models. What the rule adds is the following. Polychronakos’s Gauss law is normal ordered, so it has $s_w = 0$. Calling the effect a “level shift” is therefore a matter of which variables one uses (the Calogero coupling), not a Gauss-law effect. In particular, it should not be identified with the shifts of [1, 2], contrary to the grouping in [1] and the attribution in [8].

5 The dictionary

Reported shift	Model	Explanation in the literature	Position in the rule	Geometric meaning
$k' \rightarrow k' - N$ [1]	ADHM	normal ordering of $\varphi\varphi^\dagger$ [1, 5]	$s_{w,I} = 1, s_{w,J} = 0$	not a power of K
$k + n \mapsto k$ [2]	Jordan quiver	normal ordering of (BA) [2]	$s_w = 1$ on fundamentals	K^1 (non-equivariantly)
$k \rightarrow k + 1$ [3]	Jordan quiver, $f = 1$	reordering [3]; zero point [6, 7]; Jacobian [1]; constraint ordering [8]	adjoint $s_\varepsilon = \frac{1}{2}, s_w = 0$	overall factor, read as Calogero coupling
$\text{spin}^c / K^{1/2}$	all	half-form correction	$s_\varepsilon = s_w = \frac{1}{2}$ for all fields	level shift iff K non-trivial
gauge-covariant lattice	all	—	$s_w = 0$	no level shift

Table 1: The dictionary of quantum shifts.

6 Limits of the rule

The rule assumes that H is linear in number operators. For $H = \varepsilon N + \lambda N^2$ (the one-mode sector of $\text{tr}(B^\dagger B)^2$), a Hubbard–Stratonovich transformation evaluates the continuum coherent-state path integral exactly. For a quadratic symbol h it gives $\sum_{m \geq 0} e^{-\beta h(m+1/2)}$.

- With the Q-symbol, the energies become $E_m + \varepsilon/2 + 3\lambda/4 + 2\lambda m$. The shift depends on m , so the spectrum is distorted.
- With the Weyl symbol the shift is the constant $-\lambda/4$. This is an accident of N^2 .

The Q-symbol time lattice equals $\text{Tr}(:e^{-\delta h}:)^M$. It converges as $O(1/M)$ on a truncated Fock space. At any finite M , however, the transfer-matrix eigenvalues exceed one for $m \gtrsim 3.7M$, so the full trace diverges. This refines the statement of [14] that the time-discretised path integral does not suffer from the breakdown: the order of the limits matters. For proposed consistent continuum formulations and the ensuing debate, see [15–18]. We do not take a position on that debate. Our statement concerns only the lattice.

7 Lifting to the Procesi bundle: another regularisation ambiguity

The rank-one ADHM model computes $\chi(\text{Hilb}^n, \mathcal{O}(k))$. A natural refinement of this is the S_n -equivariant character

$$Z_P(n, k) = \sum_{|\mu|=n} \frac{T_\mu^k \tilde{H}_\mu(X; q, t)}{\prod_{x \in d(\mu)} (1 - t^{1+l} q^{-a})(1 - t^{-l} q^{1+a})} = \nabla^k h_n \left[\frac{X}{(1-q)(1-t)} \right], \quad (6)$$

which is the Frobenius series of $\Gamma(\text{Hilb}^n, P \otimes \mathcal{O}(k))$ for the Procesi bundle P . This follows from Haiman's character formula for $\Gamma(P \otimes \mathcal{B}^{\otimes l})$ [25], restricted to the S_{kn} -isotypic part (k^n) . Indeed, with $B_\mu = \sum_{(r,s) \in d(\mu)} t^r q^s$ one has $s_{(k^n)}[B_\mu] = T_\mu^k$. Its S_n -invariant and sign components are $\chi(\mathcal{O}(k))$ and $\chi(\mathcal{O}(k+1))$. In this section we ask how P can be produced by the first-order model, and find that the answer is again a question of regularisation. We first record what is known.

- Mathematically the gauge-theoretic description exists. Ginzburg's isomorphism [27]

$$(\mathbb{C}[\mathfrak{X}_{\text{norm}}] \otimes \mathbb{C}^m[V] \otimes V^{\otimes m})^{SL_n} \cong \Gamma(P \otimes \mathcal{B}^{\otimes m}),$$

restricted by Schur–Weyl duality to the isotypic part (k^n) of S_{kn} (compare [28]), says that $\Gamma(P \otimes \mathcal{O}(k))$ is the level- k Gauss-law subspace of functions on the normalised isospectral commuting variety times V . Here $\mathfrak{X}$ consists of commuting pairs together with an ordered list of their joint eigenvalues. The quantisation of P through rational Cherednik algebras is also known; see e.g. [29].

- In physics, P and $\tilde{H}_\mu$ appear as the index of nested (flag) quiver quantum mechanics [30–32]. There S_n does not act on a single Hilbert space; the Frobenius series is assembled from the flag types γ , with coefficients $\langle Z_P, h_\gamma \rangle = \chi(P^{S_\gamma} \otimes \mathcal{O}(k))$. In our conventions the $g=0$ formula of [30] is $Z_P(n, p-1)$. The virtual (supersymmetric) count deviates from χ for singular nested Hilbert schemes [31]. The virtual dimension accounts for this: it equals the actual dimension $2n$ only when every step of the flag adds one box [31, 32]. For $k=0$ the virtual counts follow from a recent theorem of Minddal [19].

Isospectral constraints. Add n points $(x_i, y_i) \in \mathbb{C}^2$ (weights t, q , permuted by S_n) and impose that they form the joint spectrum of (B_1, B_2) : $g_{ab} = \text{tr } B_1^a B_2^b - \sum_i x_i^a y_i^b = 0$ for $1 \leq a+b \leq n$. Let A^k be the span of products of k alternants (the $\det^k$ semi-invariants of the ADHM data [24, 27]), let $\Lambda = \mathbb{C}[x, y]^{S_n}$, and let $\mathcal{J} = A^1 \cdot \mathbb{C}[x, y]$ be the ideal generated by the alternants (not to be confused with the field J).

Proposition 5. *The level- k Gauss-law subspaces are as follows.*

1. *Constraints imposed scheme-theoretically (by Lagrange multipliers): $\mathcal{H}_{\text{sch}} \cong A^k \otimes_\Lambda \mathbb{C}[x, y]$.*
2. *Constraints imposed on the reduced scheme: $\mathcal{H}_{\text{red}} \cong \mathcal{J}^k \otimes \text{sgn}^k$. The same holds after normalisation.*

In both cases the S_n -invariant part is A^k , with character $\chi(\mathcal{O}(k))$.

Sketch. For (1): the generators g_{ab} are gauge invariant, and taking isotypic components for the reductive group GL_n is exact. For (2): the reduced isospectral commuting variety is irreducible, and its regular semisimple locus is dense [27]. Restricting to diagonal (B_1, B_2) with $I = (1, \dots, 1)$ is therefore injective on semi-invariants and maps $\sum_j s_j \otimes f_j$ to $\sum_j \text{alt}(s_j) f_j$. The twist sgn^k comes from the permutation matrices needed to make the restriction S_n -equivariant. $\square$

Numerically, the reduced space reproduces (6) for $n = 2$ (total degree ≤ 12), $n = 3$ (degree ≤ 10) and $n = 4$, $k \leq 1$ (degree ≤ 9). The scheme-theoretic space agrees with it only on the S_n -invariant part. For $n = 2$ the excess is

$$\mathcal{H}_{\text{sch}} - \mathcal{H}_{\text{red}} = s_{1,1} \frac{qt h_{k-1}(q, t)}{(1-q)(1-t)}, \quad h_{k-1}(q, t) = \sum_{j=0}^{k-1} t^j q^{k-1-j}, \quad (7)$$

which consists of states supported on the diagonal, such as $\det[I, B_1 I](y_1 - y_2) - \det[I, B_2 I](x_1 - x_2)$ at $k = 1$. These states are nilpotent on the scheme and vanish on the reduced scheme. We also confirmed this $n = 2$ statement with the matrix degrees of freedom kept explicitly. Imposing the constraints derivedly (by Koszul fermions over Λ) gives the Euler characteristic $\chi(\mathcal{O}(k)) Z_P(n, 0) / \chi(\mathcal{O}(0))$. This again agrees with $Z_P(n, k)$ only on the invariant part (and at $k = 0$). Thus the original partition function $\chi(\mathcal{O}(k))$ cannot distinguish the prescriptions, and their differences sit in the non-trivial S_n isotypic components. Only the reduced prescription produces $P \otimes \mathcal{O}(k)$.

Flags. Ginzburg's DG resolution [27] suggests a gauge-theoretic way to reduce: pass to a Borel or parabolic subgroup. For a composition $\gamma = (\gamma_1, \dots, \gamma_m)$ of n we take block upper triangular $B_1, B_2 \in \mathfrak{p}_\gamma$ and $I \in V$. The field J must vanish on the first $m - 1$ blocks, since that is the only way to have $IJ \in \mathfrak{p}_\gamma$. We impose $[B_1, B_2] + IJ = 0$ by Koszul fermions and treat the Levi factor $\prod U(\gamma_i)$ by a Molien integral and the nilradical by ghosts. The resulting virtual count $Z_{\text{virt}}(\gamma, k)$ has the following properties.

- For $\gamma = (1^n)$ it reduces to the residue formula

$$\langle Z_P(n, k), h_1^n \rangle = \frac{1}{((1-q)(1-t))^n} \text{CT}_z \left[\prod_{i < j} \frac{(1 - z_i/z_j)(1 - qt z_i/z_j)}{(1 - q z_i/z_j)(1 - t z_i/z_j)} \det^k h_{kn}(z^{-1}) \right], \quad (8)$$

which follows from [27, 28]. We checked it against the Atiyah–Bott sum as rational functions for $n \leq 4$.

- It equals $\langle Z_P(n, k), h_\gamma \rangle$ exactly when $\gamma = (1, \dots, 1, m)$, which is the case where the virtual dimension $n + \gamma_m + m - 1$ equals $2n$. We checked this for all compositions with $n \leq 4$ and $k \leq 1$; it is the virtual-dimension criterion of [31, 32].
- For $k = 0$ it agrees, for all compositions with $n \leq 4$, with the formula obtained from [19] with the normalisation of [31]. For $\gamma = (2, 2)$, i.e. the nested Hilbert scheme $N(1, 4, 2) = \text{Hilb}^{(4,2)}$, which is normal with rational singularities [33], the deviation is $-qt(q+t)(1+qt)/((1-q)^4(1-t)^4(1+q)^2(1+t)^2)$. The deviation therefore comes from excess dimension, not from non-reducedness.

The literature treats $k = 0$ (or, in [31], $k = -1$) only. Our computations suggest a closed form for all k :

$$Z_{\text{virt}}(\gamma, k) = \left\langle Z_P(n, k), h_{\gamma_m} \prod_{i < m} \frac{h_{\gamma_i}[(1-qt)X]}{1-qt} \right\rangle. \quad (9)$$

Equivalently, at each fixed point ν the sum over the tails of the flag equals $\langle \tilde{H}_\nu, h_{\gamma_m} \prod_{i < m} h_{\gamma_i}[(1-qt)X]/(1-qt) \rangle$, independently of k . Since $h_1[(1-qt)X]/(1-qt) = h_1$, the deviation vanishes exactly in the one-box-step case. We checked (9) for all compositions with $n \leq 4$ and $k \leq 2$, and for seven compositions with $n = 5$ and $k \leq 1$ (total degree $\leq 8-10$). Independently, the fixed-point values tabulated in [31] for $N(1, 4, 2)$ coincide with $\langle \tilde{H}_\nu, h_2(h_2 - qt e_2) \rangle$ for all five ν . We have not attempted a proof.

In summary, the Procesi bundle is reached by the first-order model only through a specific treatment of a non-complete-intersection constraint, either by reduction or by a flag resolution

with one-box steps. The two natural alternatives, scheme-theoretic and derived imposition, differ from it in a controlled way that is invisible in $\chi(\mathcal{O}(k))$. This extends the theme of the previous sections from level shifts to S_n -refined characters.

8 Conclusions

One rule with two exponents per field organises the quantum shifts of first-order gauged matrix models. Part of the resulting dictionary confirms identifications already made by the original authors [1, 2, 6, 7]. Its use is to make the remaining distinctions sharp and to resolve conflicting attributions. It separates three kinds of shift: Gauss-law (dictionary) shifts (s_w), zero-point and Calogero-type shifts (s_ε), and genuine spin^c twists ($s_\varepsilon = s_w = \frac{1}{2}$). The comparison between trivial and non-trivial canonical bundles is essential: on Hilb^n several inequivalent prescriptions cannot be told apart.

The same theme reappears for the Procesi bundle: the S_n -refined character depends on how a non-complete-intersection constraint is imposed, although the unrefined partition function does not. A natural next step is a phase space with both non-trivial K and a holomorphic constraint, together with non-linear Hamiltonians in the gauged setting, and a proof of (9).

A Remarks on JK residues and wall crossing

For rank-one ADHM we also verified the following.

- The fixed-contour Molien integral selects the chamber $\text{sign}(\zeta) = \text{sign}(k)$, consistently with $\zeta = k'$.
- The wall-crossing difference is

$$Z_+ - Z_- = \chi(\mathcal{O}(k)) - (qt)^{-(k+1)n} \chi(\mathcal{O}(k))^\vee. \quad (10)$$

It vanishes at $k = 0$ and factorises at $k = 1$ because $\chi(\mathcal{O}(1)) = \chi(\mathcal{O}(-1))$, which we observed for $n \leq 5$.

- In rank two there is no wall crossing for $|k| < N$.

These follow from the general results of [20, 22]. A case analysis of the background charge already appears in [23].

B Scope of the numerical checks

All checks except those marked otherwise are automated tests (429 in total). Power series in q, t are compared up to total degree 10–16, and numerical integrals are compared to relative accuracy 10^{-4} – 10^{-12} . The ranges are:

- rank-one ADHM: $n \leq 4$, $0 \leq k \leq 3$; JK residues for $n \leq 5$;
- lattices: $n \leq 3$;
- rank two: $n \leq 2$;
- Grassmannians: $n \leq 2$, $N \leq 4$;
- Jordan quiver: $n \leq 3$, $f \leq 2$;
- isospectral constraints (Section 7): $n = 2$ with $k \leq 3$, $n = 3$ with $k \leq 2$ and $n = 4$ with $k \leq 1$, by linear algebra modulo a prime below 2^{23} (total degree 9–12); for $n = 2$ also with explicit matrices, exactly over $\mathbb{Q}$ ($k \leq 2$, total degree ≤ 6);

- flags: residue formula exactly as rational functions for $n \leq 4$; virtual counts as series up to total degree 8–10 for all compositions with $n \leq 4$, $k \leq 2$, and for seven compositions with $n = 5$, $k \leq 1$ (the latter outside the automated tests).

For $k \geq 2$, the Hilbert series of products of alternants is computed probabilistically, from ranks modulo a large prime.

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